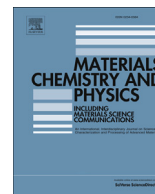




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Materials Chemistry and Physics

journal homepage: www.elsevier.com/locate/matchemphysSelective dissolution of an amorphous $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy in organic acids and dilute HCl solutionFengxiang Qin ^{a,*}, Zhenhua Dan ^{b,c,d,**}, Nobuyoshi Hara ^e, Weirong Li ^f, Yangde Li ^f^a School of Materials Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China^b College of Materials Science and Engineering, Nanjing TECH University, Nanjing 210009, China^c The Synergetic Innovation Center for Advanced Materials, Nanjing Tech University, Nanjing 210009, China^d Institute for Materials Research, Tohoku University, Sendai 9808577, Japan^e Department of Materials Science, Tohoku University, Sendai 9808579, Japan^f Dongguan Eontech Co., Ltd, Dongguan 523662, China

H I G H L I G H T S

- Nanoporous copper can be obtained in organic acids and it does not in dilute HCl.
- Nanoporous copper has a size of nanopores and ligaments less than 20 nm.
- Formic and acetic acids have a buffer ability to prevent the passivation.
- Low mobility and large size of organic acids retard the surface diffusion of Cu adatoms.

A R T I C L E I N F O

Article history:

Received 11 September 2015

Received in revised form

29 April 2016

Accepted 1 May 2016

Available online xxx

Keywords:

Amorphous materials

Microporous materials

Etching

Corrosion

A B S T R A C T

Selective dissolution behavior of an amorphous $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor alloy in organic acids has been investigated. Nanoporous copper (NPC) with a pore size of 11–19 nm and ligament of 16–19 nm is obtained after dealloying in 0.01 M formic and acetic acids. NPC formed in formic acid has a larger pore size and similar ligament size than that in acetic acid. Plot of nanopore size versus dealloying time demonstrates the formation of NPC is controlled by the surface diffusion of Cu adatoms. Low diffusion ability of anions with more C–H chains from organic acids results in the formation of finer NPCs. On the other hand, NPC is unable to form in 0.01 M HCl solution and corrosion products consist of Y_2O_3 , $\text{Y}(\text{OH})_3$, $\text{Mg}(\text{OH})_2$ and metallic Cu. The thick corrosion products passivate the surface to prevent from the further aggregation of corrosion. The formation of the finer NPC in organic acids benefits from slow surface diffusion of Cu adatoms, the buffer ability of the organic solutions and the low mobility of the organic molecule.

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1. Introduction

Nanoporous metals (NPMs) have attracted extensive attention in many research fields since they have advantages in various applications, such as catalysts, sensors and actuators [1–5]. The

chemical or electrochemical methods including selective dissolution of less noble elements or phases in electrolytes, so-called dealloying [3,5–10] have used to fabricate NPMs. Among the methods mentioned above, chemical/electrochemical dealloying is regarded as a facile method for preparing NPMs. Fabrication of nanoporous Au, Au and Cu have been extensively investigated in many aspects (i.e. initial formation of nanoporosity [11–14], effect of the initial microstructure of precursors [15–19], dealloying solutions containing inorganic acids [3,6,8,9,11–19] and/or inorganic acids [20,21] or neutral salt solutions [10]). Amorphous precursor alloys, such as Cu–Hf–Al [16], Ti–Cu [8,17,22], Mg–Cu–Y [23] and Pd–(Cu)–Ni–P [24,25] alloys are good for forming a uniform nanoporous microstructure because they are homogenous in chemical

* Corresponding author. School of Materials and Science and Engineering, Nanjing University of Science and Technology Xiaolingwei 200, Nanjing, Jiangsu 210094, China.

** Corresponding author. College of Materials and Science and Engineering, Nanjing TECH University, Xinmofan Road, No. 5, Nanjing, Jiangsu 210009, China.

E-mail addresses: fengxiangqin@njut.edu.cn (F. Qin), zhenhuadan@njtech.edu.cn (Z. Dan).

¹ Equal contribution to the present paper.

composition and microstructure and without defects such as grain boundary or intermetallics. The selection of the amorphous $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor alloy can weaken the effect of the initial microstructure and local frustration of chemical composition on final nanoporosity, and the selective dissolution behavior in organic acidic solutions or inorganic acids can be clarified more simply. So far the dealloying behavior of multi-component alloys in organic acids is still unclear. Many researches also demonstrate that the properties of dealloying solutions influence the dealloying behavior, such solution pH [8,26,27], addition of macromolecular organics [22,23], and dealloying temperature [6,7,28,29]. Dealloying behavior of Ti–Cu and Mg–Cu–Y amorphous alloys in PVP-added acids have been reported recently, and the refinement of the nanoporous copper (NPC) has been refined to a level with a pore size around 10 nm [22,23] by introducing macromolecular organic chemicals to restrict the mobility of Cu adatoms in inorganic acids. The experimental results have been reported to demonstrate the dealloying behaviors of Mn–Cu alloy in acetic acids which is predicted to be different from that in inorganic acids since the concentration of acetic acid greatly affects its competition between association and ionization of acetic molecules [30]. Neutral macromolecules restrict the diffusion, and organic acids have an ability to maintain their solution pH. Generally speaking, organic acids have a large molecular size and lower mobility in aqueous condition, and higher pH than inorganic acids [31]. Since there are many unclear points in the dealloying behaviors in organic acids, it is worth to make more investigation in various organic acids. The finer nanoporous structure is highly expected to obtain in organic acids after dealloying, and the mechanism of different dealloying behaviors should be addressed.

In the present study, typical formic and acetic acids with single and double C–H chains had been chosen to be dealloying solutions. The dealloying behaviors of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ amorphous alloys in these two solutions had been investigated through comparing with that in hydrochloric acid from aspects of microstructure of nanoporous alloys, surface morphology and corrosion products.

2. Experimental procedure

A Cu–Y pre-alloy was produced by arc-melting pure Cu (99.99 wt%) and Y (99.9 wt%) under a Ti-gettered argon atmosphere. A ternary $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy was prepared by re-melting the Cu–Y pre-alloy with high purity Mg (99.9 wt%) in a high-frequency induction furnace under a high-purity argon atmosphere. Amorphous ribbons with a thickness of 30 μm and a width of 5 mm were obtained from this alloy by melt-spinning on a rotating copper roller.

Dealloying was performed for 0.9–7.2 ks in 0.01 M HCl solution, formic acid and acetic acid (Sigma-Aldrich) under free corrosion condition. The crystalline state of the as-spun and dealloyed ribbons was confirmed by an X-ray diffractometer (Rigaku, Smart-Lab) and a transmission electron microscope (JEOL, HC2100). An ion milling method was used to prepare the TEM samples. The surface morphology of the dealloyed ribbons was observed by a scanning electron microscope (JEOL, FIB4610). The corrosion products and residual phases of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after immersing in 0.01 M HCl and formic acid for 7.2 ks were confirmed by an X-ray photoelectron spectroscopy (Shimadzu Kratos, AXIS-Ultra DLD) with a monochromatized Al K α radiation source (1486.6 eV) and an energy dispersive X-ray spectroscopy (JEOL, FIB4610). The size of nanopores and ligaments was obtained by measuring over 150 sites on the SEM morphology by using single-chord method.

3. Experimental results and discussion

3.1. Crystalline properties of as-spun and dealloyed $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys

Fig. 1 shows the XRD patterns of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbons before and after dealloying in 0.01 M organic acid solutions for 7.2 ks. A strong broad diffraction peak around 36.3° appeared in the XRD pattern of as-spun $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon, which is typical for the amorphous alloys (Fig. 1a). The diffraction pattern of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon after dealloying in 0.01 M HCl solution for 7.2 ks is shown in Fig. 1b. The series of diffraction peaks were assigned to $\text{Mg}(\text{OH})_2$, Cu_2O and Cu according to diffraction patterns of $\text{Mg}(\text{OH})_2$ (JCPDS card No.: 74-2220), face-centered cubic (fcc) Cu powder (JCPDS card No.: 02-1225) and Cu_2O powder (JCPDS card No.: 74-1230), indicating that the complex corrosion products formed on $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon after immersion in HCl solution. It is hard to tell that the Y-containing compounds such as $\text{Y}(\text{OH})_3$ or Y_2O_3 existed from XRD pattern since many diffraction peaks overlapped with those from Cu or Cu_2O and $\text{Mg}(\text{OH})_2$. As shown in Fig. 1c and d, the diffraction peaks at $2\theta = 43.1^\circ$, 50.4° and 74.1° from dealloyed ribbons in formic acid and acetic acid were assigned to Cu (111), (200) and (220), respectively. The slight shift in the diffraction peaks to lower angles than the standard data for Cu indicates that the lattice constants became larger. The calculated lattice constants of the residual phase after dealloying the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon in 0.01 M formic and acetic acid solution were 0.36292 nm and 0.36296 nm, respectively. The other small peaks at $2\theta = 29.5^\circ$, 36.4° and 61.4° were attributed to Cu_2O (110), (111) and (220). The results of the XRD analysis suggest that the residue of dealloyed alloys formed in organic acids is fcc Cu with a trace amount of Cu_2O and the corrosion products in HCl solution are composed of $\text{Mg}(\text{OH})_2$, Cu_2O and fcc Cu, and others unable to be identified by XRD. The lattice constants of NPCs are slightly larger than that of standard value of Cu, indicating that some defects, such as twin boundary or dislocations involved into Cu lattices during the rearrangement of Cu adatoms.

3.2. Passivation of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon in dilute HCl solution

The corrosion morphology of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon in 0.01 M HCl solution after immersion of 900 s and 7.2 ks is shown in Fig. 2. There

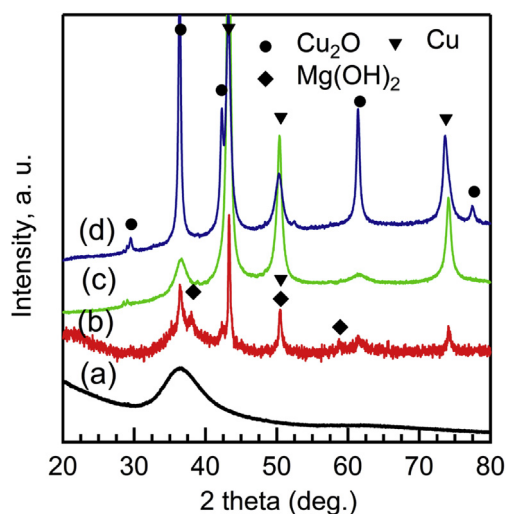


Fig. 1. XRD patterns of as-spun $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy (a) before and after dealloying in 0.01 M HCl (b), formic acid (c) and acetic acid (d) for 7.2 ks.

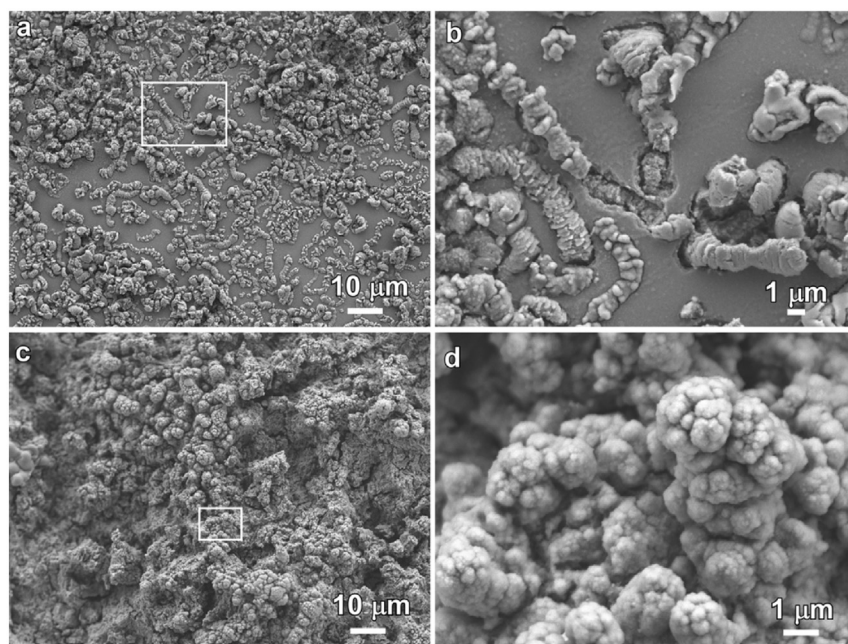


Fig. 2. Surface morphology of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after immersing in 0.01 M HCl solution for 900 s (a) and 7.2 ks (c). Images b, d are magnified images in frames in a, c.

were many bubbles formed on the surface of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys during immersion in 0.1 M HCl solution. This fact indicates that the fast dissolution behavior of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys in 0.1 M HCl solution took place. The similar behavior has been observed when $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys are immersed in 0.1 M H_2SO_4 solution before [23]. Although the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys suffered severely from the local attack by Cl^- ions in 0.01 M HCl condition at the beginning of immersion of 900 s as shown in Fig. 2a, the corrosion products gradually covered all over the surface with the increase of the immersion time showing in Fig. 2c. After immersion of 7.2 ks, the inner part of ribbons still remained. As indicated in Fig. 1b, the broad diffraction peaks around 36.3° in Fig. 1b indicated that there were still some amorphous precursor alloys existing. After immersion of 900 s, there were many worm-like corrosion products formed on the surface of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbon at some local sites. In some regions, the surface seemed remaining uncorroded. The corrosion mainly went beneath the surface oxide layer in the form of filiform corrosion in Fig. 2b. According to the obtained XPS analysis data of as-spun $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy, surface oxide layer mainly consisted of $\text{Mg}(\text{OH})_2$ (66.4 at.%), MgO (16.0 at.%), Y_2O_3 (5.5 at.%) and metallic Mg (3.4 at.%), Cu (3.3 at.%), Y (5.3 at.%). At the initial immersion, Cl^- ions partially attacked the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ substrate in the form of filiform corrosion. The generated corrosion products partially covered the surface and the rest regions were still under attack by Cl^- ions (Fig. 2a). The corrosion products were analyzed by EDX showing in Fig. 4a and b. EDX spectra indicates that the corrosion product layer mainly consisted of Mg, Y, Cu and O. With an increase of immersion time, the corrosion products accumulated on the surface and expanded along the surface (Fig. 2c). XPS spectra of corrosion products on $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor after immersion of 7.2 ks in 0.01 M HCl solution are shown in Fig. 8. The presence of the peak at 46.1 eV in Mg 2p spectrum indicated that the Mg mainly existed in the form of $\text{Mg}(\text{OH})_2$. The subpeak at 936.6 eV in Cu 2p was assigned to $\text{Cu}(\text{OH})_2$ compound [32], and the secondary subpeak at 932.5 eV is considered to rise from the Cu^{m} or Cu_2O . The Y 3d_{3/2} and 3d_{5/2} doublet is the most prominent XPS spectra for Y (Fig. 8c). Doublet peaks at 157.9 and 159.9 eV was assigned to Y_2O_3 [33–35]. The Y 3d doublet at 159.5

and 161.5 eV of the breakdown film is attributed to Y hydroxide (YOOH and/or $\text{Y}(\text{OH})_3$) [36–38]. Fig. 8d shows two O 1s peaks at 529.1 and 531.2 eV assigned to oxide and hydroxide in the corrosion product layer. The coverage of the corrosion products increased with the immersion time as shown in Fig. 2, and the corrosion products piled up to form the morphology as shown in Fig. 2c and d. As listed in Table 1, the corrosion products were mainly composed of Y-containing oxides (16.2 at.%), such as $\text{Y}(\text{OH})_3$ and Y_2O_3 , after immersing in 0.01 M HCl solution for 7.2 ks. The CuO and Y_2O_3 formed on $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys [33] incorporated in the corrosion product layer via substitution of the magnesium cations and a subsequent increase in the local positive charge. The formation of abundant yttrium oxide and yttrium hydroxide indicated that the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor was passivated by Y^{3+} ions.

3.3. Selective dissolution of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy in organic acids

Corrosion potential of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy was measured to be -1.02 ± 0.05 V in formic acid, -0.98 ± 0.05 V in acetic acid and -1.42 ± 0.05 V in 0.01 M HCl. $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor alloy was in an active dissolution state in three organic acids. On the other hand, the nanoporous structures were obtained after dealloying $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor as shown in Fig. 3. Bi-continuous nanoporous structure with an open-cell nanopores and ligaments formed after dealloying under free immersion conditions. The nanoporous structure formed in formic acid and acetic acid was similar, while the morphology was much different from those in 0.01 M HCl solution. The change of nanopore size, d , and ligament size, l , is shown in Fig. 5. The NPCs formed in 0.01 M formic acid after dealloying for 900 s, 1.8 ks, 3.6 ks and 7.2 ks had a ligament size of 17.3 nm, 16.1 nm, 15.9 nm and 16.4 nm and a pore size of 11.9 nm, 13.4 nm, 14.7 nm and 18.6 nm. In comparison to those dealloyed in formic acid, the average size of nanopores and ligaments of NPCs formed in acetic acid after an immersion of 7.2 ks was confirmed to be 16.2 nm and 17.1 nm, respectively. The NPCs formed in both formic acid and acetic acid are considered to have a similar nanoporosity after dealloying in organic acids for 3.6 ks since nanopores and ligaments had a similar average size.

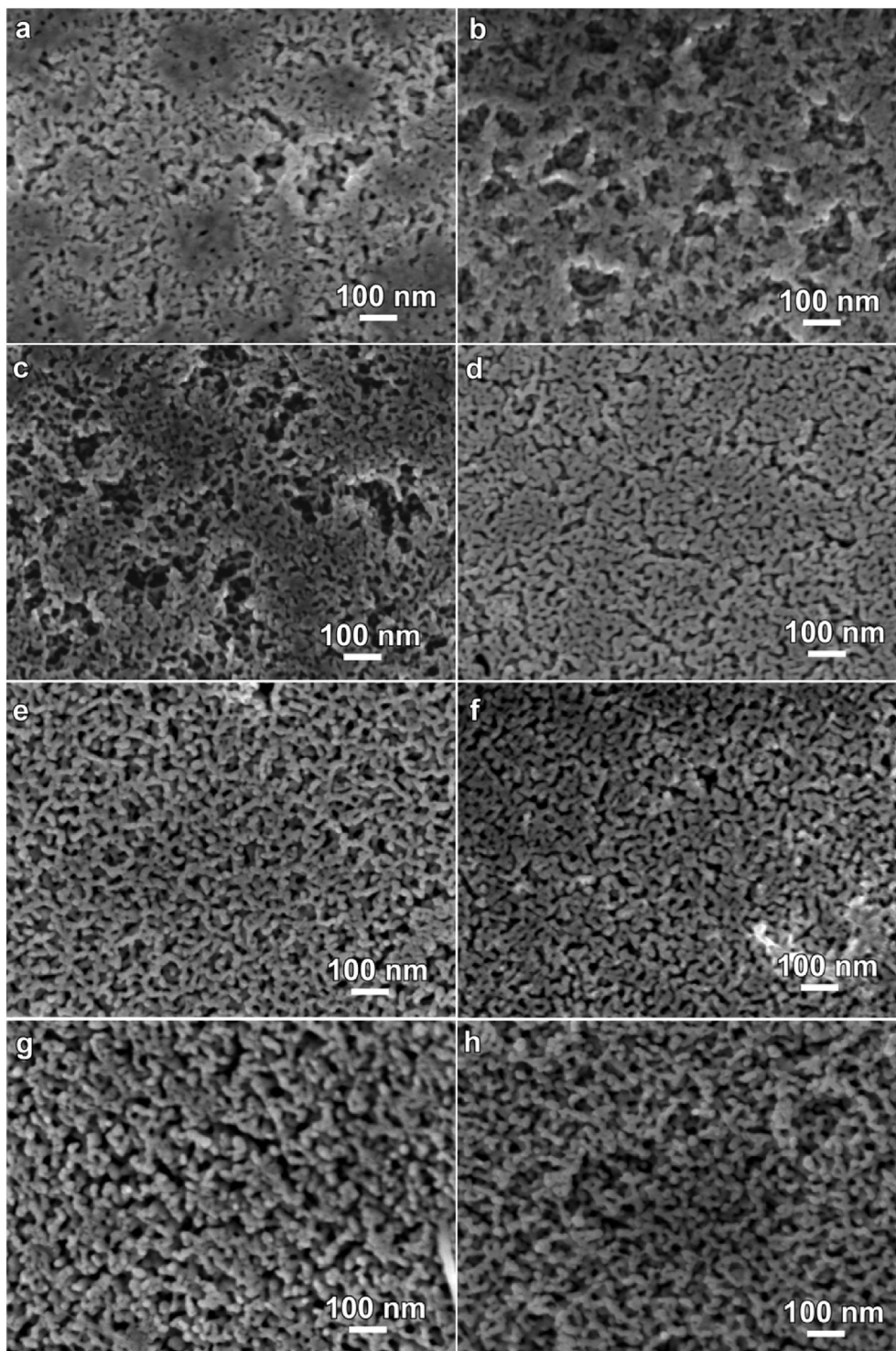


Fig. 3. Surface morphology of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after dealloying in 0.01 M formic acid (a, c, e, g) and 0.01 M acetic acid (b, d, f, h) for 900 s (a, b), 1.8 ks (c, d), 3.6 ks (e, f) and 7.2 ks (g, h).

The solid-liquid interface diffusion mechanism has been widely proposed to explain the nucleation and growth process of ligament of NPMs during dealloying of both crystalline and amorphous precursor alloys [29,39–41]. Thus, referring to the isothermal grain growth in polycrystalline materials [42], the interplay between ligament size of NPC and electrochemical dealloying conditions, i.e. leaching time t and temperature T , can be modeled with the following equation [29]:

$$d^n = KtD_s$$

where D_s is the surface diffusivity of Cu atoms at a certain temperature, K is a constant, t is the time, and n is the coarsening exponent.

The coarsening exponent n can be directly measured by plotting the $\ln d$ vs $\ln t$ curves, as shown in Fig. 5b. The excellent linear

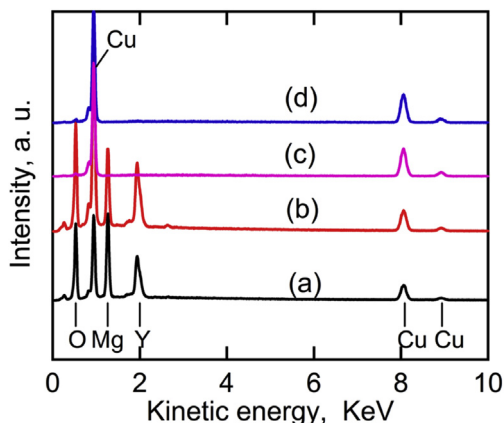


Fig. 4. EDX spectra of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after dealloying in 0.01 M HCl for 900 s (a) and 7.2 ks (b) and in 0.01 M formic acid (c) and 0.01 M acetic acid (d) for 7.2 ks.

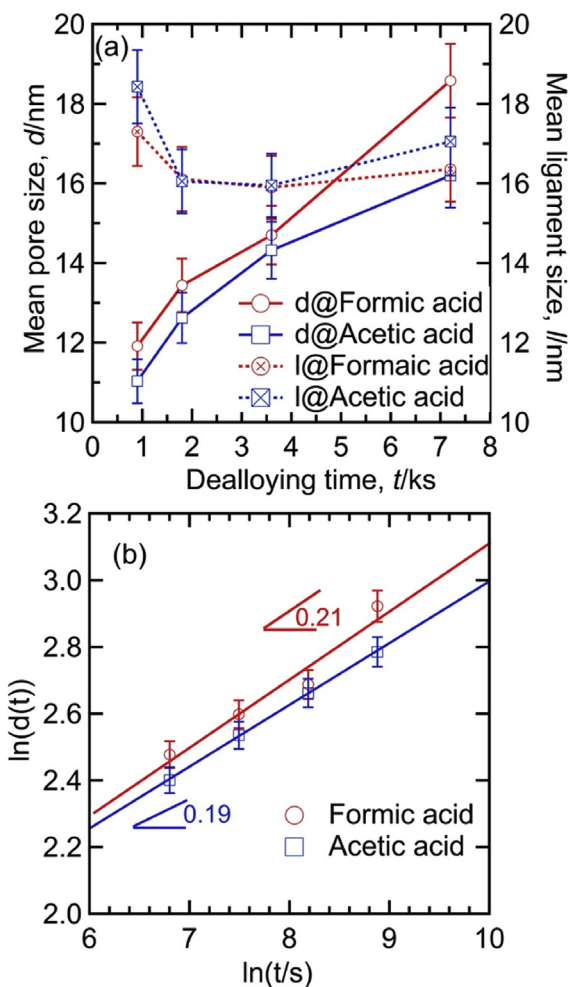


Fig. 5. Mean pore and ligament size vs. dealloying time (a) and plot of $\ln(d(t))$ vs. $\ln(t)$ (b) of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after dealloying in 0.01 M formic and acetic acid.

relations between $\ln d$ and $\ln t$ further verify the logarithmic coarsening mechanism of NPC in organic acid solutions. The nearly identical slope, 0.21 in formic acid and 0.19 in acetic acid, of the fitting lines for $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ amorphous precursors implies an invariable coarsening mechanism in different solutions. The n value for the coarsening of NPC is determined to be 4.8 in formic acid and

5.3 in acetic acid, which is very close to the kinetic parameter, 4, reflecting surface relaxation of roughened metals in solutions [29,43]. The facts presented above indicates that the formation of the NPCs is controlled by the surface diffusion of the Cu adatoms.

The nanoporous structure formed in organic acids was confirmed by TEM shown in Figs. 6 and 7. As shown in Fig. 6, the nanoporous structure had a bi-channel structure with Cu ligament and nanopores. The diffraction pattern obtained in a wide region shows the typical diffraction rings of the nanocrystallinity of the residual compounds. That in the small regions had a crystalline pattern with various diffraction matrix. As shown in Fig. 6c, the nanodiffraction pattern could be assigned to Cu [112] and [011]. The Cu ligaments consisted of fcc Cu crystals. It is worth noting here that the minor Cu_2O compound coexisted with Cu ligament, specially at the outer surface of the Cu ligaments. The coexistence of Cu_2O and Cu was also confirmed by XRD and EDX macroscopically. The elemental distribution of the Cu and O is shown in Fig. 7 a and b. As has been described above, Cu_2O compound tends to form at the outside of the Cu ligaments. Since the TEM samples were prepared by ion-milling slicing method, it is considered that the Cu_2O compounds were formed due to the surface oxidation of Cu ligaments when they were exposed to the atmosphere.

The XPS spectra of Mg 2p, Cu 2p $_{3/2}$, Y 3d $_{5/2}$, O 1s, C 1s and wide scan for dealloyed $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbons in Fig. 8 (II) show different chemical composition of the residual compounds after immersing in organic acids from that in 0.01 M HCl solution (I). The Mg and Y spectra demonstrates that the Mg and Y were selectively dissolved from the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor and the residual compound was confirmed to be metallic Cu since the presence of the single strong peak at 932.3 eV assigned to $\text{Cu}^{\text{m}}/\text{Cu}_2\text{O}$. The shape of the C 1s spectrum in Fig. 8 e(II) differed from that lower one. The sub-peaks at 284.8 eV, 286.3 eV and 288.2 eV were assigned to C–C/H bonding, C–O bonding and C=O bonding, respectively. In the spectra of C 1s in Fig. 8 e(II), one large sub-peak centered at 288.2 eV was assigned to –COO bonding and its presence is considered to be due to the adsorption of HCOOH molecule. The wide scans demonstrate that the Cu peak at 932.3 eV in Fig. 8 f(II) is much higher than that obtained in HCl solution. It is worth stating that the EDX spectra in Fig. 4 c and d and the XRD pattern in Fig. 1 c and d demonstrated that the residual compounds in organic acids were fcc Cu and trace Cu_2O which differed from the corrosion products formed in 0.01 M HCl solution. The chemical composition of the corrosion products formed in 0.01 M HCl solution and other organic acids is listed in Table 1. The main chemical composition of corrosion product layer in inorganic acid was composed of $\text{Cu}(\text{OH})_2$, metallic Cu, Y_2O_3 , $\text{Y}(\text{OH})_3$ and $\text{Mg}(\text{OH})_2$ and their hydrated chemical compounds. On the other hand, the residual compounds after dealloying in organic acids are confirmed to be metallic Cu, and some trace $\text{Y}(\text{OH})_3$. The drastic difference in the chemical composition demonstrates that the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy can be dealloyed to form Cu ligament in organic acids and it cannot be achieved in dilute HCl solution.

3.4. Discussion

According to the FACTSAGE database, $\text{Mg}(\text{OH})_2$ and $\text{Y}(\text{OH})_3$ are able to form while solution pH is higher than 11.4 and 10.4. This fact states that the solution pH increased faster when $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys were immersed in 0.01 M HCl solution in comparison to that in organic acids. The native oxide layer mainly consisted of $\text{Mg}(\text{OH})_2$ and Y_2O_3 species. $\text{Mg}(\text{OH})_2$ (brucite) has a hexagonal crystal structure and undergoes easy basal cleavage causing cracking and curling of the films. The density of the $\text{Mg}(\text{OH})_2$ film is higher than that of magnesium, therefore, compressive rupture of the films takes place resulting in continual exposure of fresh metal surface.

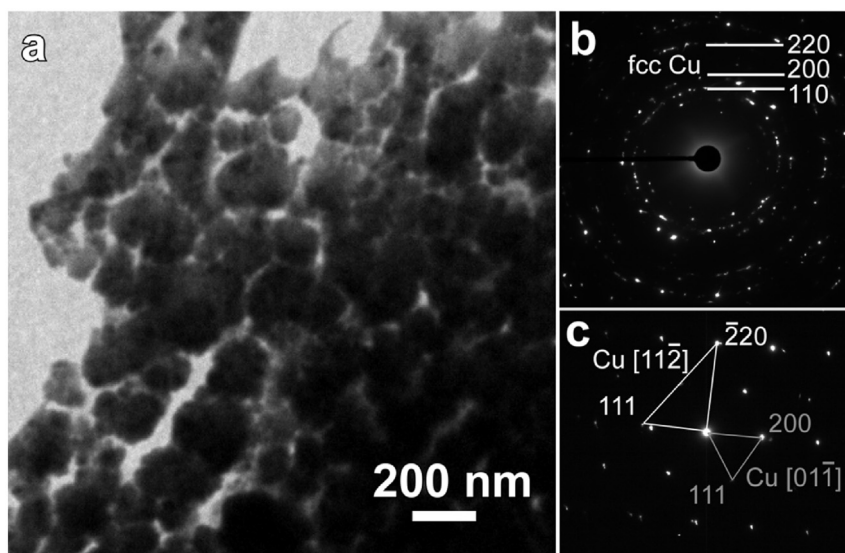


Fig. 6. Low magnification bright field TEM image (a), selective area diffraction pattern at a wide area (b) and in a small area (c) of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after dealloying in 0.01 M formic acid for 7.2 ks.

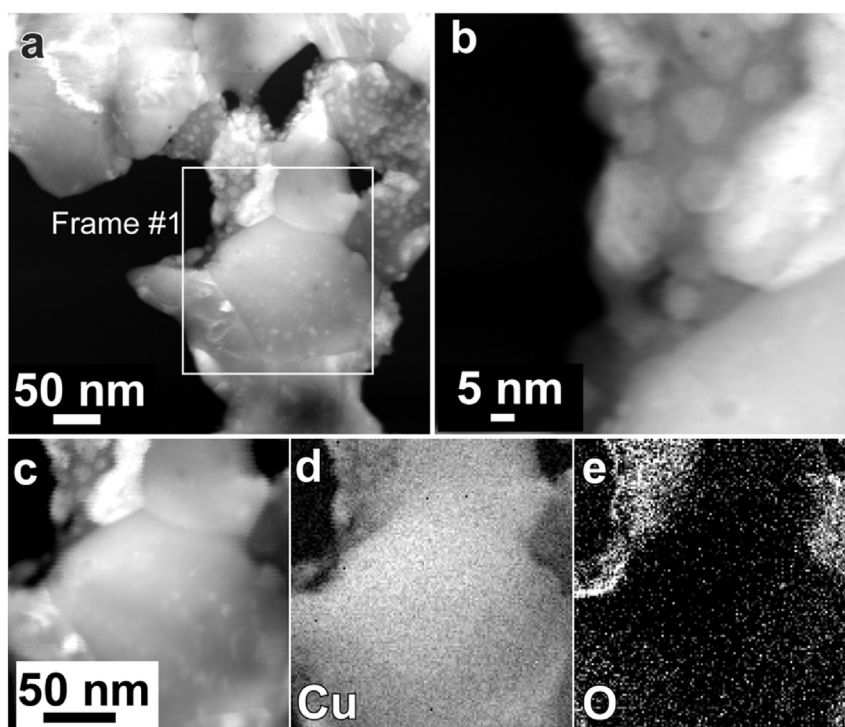


Fig. 7. Dark field TEM image (a) and high resolution dark field TEM image (b) and the magnified dark field TEM image in Frame #1 in a (c) and the corresponding element mapping of Cu (d) and O (e) of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after dealloying in 0.01 M formic acid.

When chloride ions are present in the electrolyte, the corrosion attack is even more severe due to the easy penetration of the hydroxide films by Cl^- ions [44]. As shown in Figs. 2 and 3, the chloride ions and hydrogen ions have a strong ability to attack the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloys at the initial immersion locally. The mobility of Cl^- ions has been reported to be high in many researches, such as the corrosion attack of stainless steel in HCl solution [45], the atmospheric corrosion of aluminium alloys in Cl^- -containing tropical environment [46]. The investigation of the corrosion behavior of pure Mg and Mg–Al–Zn alloys in 0.1 M NaCl solution demonstrates

that (1) thick $\text{Mg}(\text{OH})_2$ surface films suffer local breakdown in the presence of Cl^- and SO_4^{2-} ions; (2) these materials undergo localized corrosion such as pitting and filiform corrosion under free immersion condition since the open-circuit potential exceeds their breakdown potential on $\text{Mg}(\text{OH})_2$ -covered alloys [44]. In general, amorphous precursor alloys may have better corrosion resistance in comparison to their counterpart crystalline alloys since the amorphous alloys are homogeneous in chemical composition, no defects such as grain boundary, intermetallic phases acting as initial sites of corrosion, and highly uniform in microstructure [47,48].

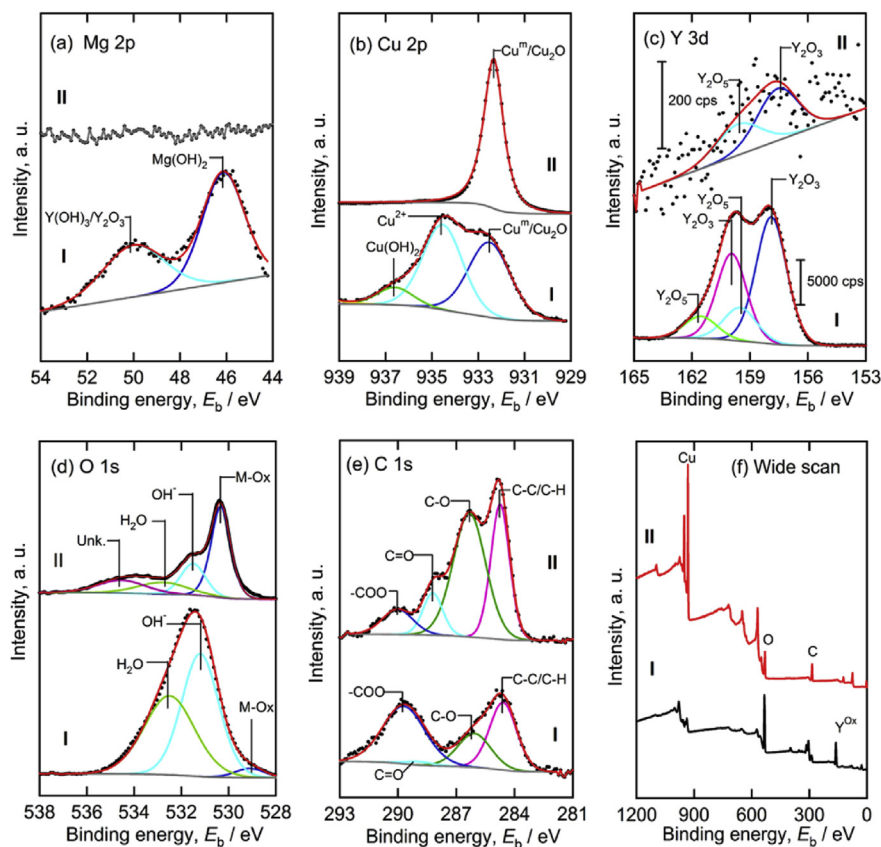


Fig. 8. XPS spectra of Mg 2p (a), Cu 2p (b), Y 3d (c), O 1s (d), C 1s (e) and wide scan (f) of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after immersing in 0.01 M HCl (I) and 0.01 M formic acid (II) for 7.2 ks.

Table 1

Chemical composition of surface layer on $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ alloy after immersing in 0.01 M HCl and formic acid for 7.2 ks (Unit: at.%).

Solution	$\text{Cu}^{\text{m}}/\text{Cu}_2\text{O}$	Cu^{ox}	Mg^{ox}	Y^{ox}	Cl^-	OH^-	M–O	H_2O	Unkn.
HCl	1.9	3.6	3.2	16.2	0.8	38.5	2.3	33.5	–
Formic Acid	43.7	–	–	0.2	–	11.5	26.8	8.7	9.1

Amorphous $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ ribbons shows significantly improved corrosion resistance in strongly alkaline electrolyte compared to pure Mg and $\text{Mg}_{79}\text{Cu}_{21}$ [49,50]. The passivation of $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ amorphous ribbons in 0.01 M HCl solution was caused by the formation of other oxides such as CuO , Y_2O_3 and $\text{Y}(\text{OH})_3$ in main compounds of MgO and $\text{Mg}(\text{OH})_2$ to give a continuous and uniform coverage of the alloy surface. It is well known that the cations of rare earth elements, such as Ce^{4+} and Y^{3+} , have an ability to retard or even passivate the surface [49–51]. The passivation of the surface can be triggered by incorporation of Cu and Y for $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ solutes in (1) the hexagonal MgO lattice to prevent incorporation by diffusion of Cl^- anions; (2) the MgO lattice to increase its volume ratio with respect to Mg, so compensating for or counteracting internal stresses, reducing susceptibility to cleavage and avoiding ionic diffusion paths [52–55]. The thick corrosion product layers consisting of $\text{Mg}(\text{OH})_2$, Y_2O_3 , $\text{Y}(\text{OH})_3$, and CuO or Cu prevent the penetration of Cl^- ions from the further corrosion attack. Since Y^{3+} ions generated from the selective dissolution of Mg and Y atoms from $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ precursor passivated the exposed front of the kinks or terrace on $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ surface [49] and the corrosion product layer composed of compounds with a low solubility in aqueous solutions protects the $\text{Mg}_{65}\text{Cu}_{25}\text{Y}_{10}$ surface from the further attack by Cl^- ions. As a result, the formation of Cu clusters

were blocked by the corrosion product layer.

The secondary dielectric properties of organic acids somewhat make them have a buffer ability to maintain pH at a stable level. It is worth noting that the organic acids commonly have a relatively lower dielectric constant, pK_a [31]. For instance, formic acid ($\text{H}-\text{COOH}$) and acetic acid (CH_3-COOH) have pK_a of 3.77 and 4.76, respectively. It is certain that the molecular size increased with the increase of the C–H chains. The diffusion of metal adatoms is expected to be retarded more efficiently while organic acid molecules with more C–H chains are introduced. The values of pore-to-ligament ratio (d/l) of NPC formed in acetic acid changed from 0.60 to 0.95, and that in formic acid did from 0.69 to 1.14, as shown in Fig. 5a. The ratio in acetic acid was smaller than that in formic acid, indicating that the surface diffusion of Cu adatoms was slightly suppressed by the introduction of the organic acids with larger molecule. The formic and acetic acids chosen in the present study have close values of pK_a and similar molecular structure, limited effects on the refinement of NPC are anticipated. As indicated in the former researches, the suppression effects of large-sized PVP in dealloying inorganic acids on the formation of NPC are much stronger than that in the formic and acetic acids [22,23]. As well known, ionic liquids are composed of large-sized ions which result in low mobility even high viscosity. In some extent, ionic liquids have similar effects on formation of ultrafine NPMs as these organic acids. However, the low mobility and activity of ions in ionic liquids play some important roles in the formation of NPMs [56–58]. On the other hand, the formation of NPC was mainly attributed to the low dielectric constant, pK_a , and high buffer ability of organic acids. As well known, HCl acid has an ability to ionize completely in aqueous solutions. Although organic acids are ionized partially and have a higher pH value than HCl, the gradual

release of H^+ ions results in a stable solution pH, and the formic and acetic acidic solutions have a buffer ability. As has been reported before, the formation of nanoporous Ni has been achieved due to the buffer ability of $NH_4H_2PO_4$ solution [27]. The similar researches related to nanoporous Au and Cu have been reported to demonstrate the effect of the buffer ability of dealloying solutions on the formation of nanoporous structure [10,26,39,40]. As indicated by Cu–H₂O diagram, Cu can maintain its metallic state when pH is lower than 4.6 [59]. On the basis of data presented in Figs. 1, 4, 6 and 8, NPC was formed as solution pH lower than 4.6. That is to say that the pH of the organic acidic solutions was maintained in the narrow range of 3.8–4.6. The organic acids (i.e. HCOOH, CH₃COOH) exhibited their strong buffer ability in comparison to inorganic acids (i.e. HCl). It can be predicted that the Y^{3+} and Mg^{2+} cations cannot incorporate OH[–] anions to passivate the surface and stop the upcoming selective dissolution reaction. In a word, the formation of NPC benefits from the strong buffer ability of organic acids.

4. Conclusion

The behavior of selective dissolution of an amorphous $Mg_{65}Cu_{25}Y_{10}$ precursor alloy in the organic acids (HCOOH, CH₃COOH) and inorganic acid (HCl) undergoes differently.

- (a) In organic acids
 - 1) The corrosion products are composed of $Mg(OH)_2$, $Y(OH)_3$, Y_2O_3 and trace Cu_2O or Cu and passivate the surface to stop the continuous selective dissolution of Mg and Y elements.
 - 2) The nanoporous structure has not been obtained.
- (b) In organic acids
 - 1) Bi-continuous NPC with a pore size of 11–19 nm and ligament size of 16–19 nm is formed after dealloying.
 - 2) NPC formed in formic acid has a larger pore size and similar ligament size than that in acetic acid. Low diffusion ability of anions with more C–H chains from organic acids results in the formation of finer NPCs.
 - 3) The formation of NPC is controlled by the surface diffusion of Cu adatoms, and this process is improved by the organic acids.
 - 4) Formation of fine NPC benefits from the buffer ability and low mobility of the organic molecule.

Acknowledgment

This work is financially supported by National Natural Science Foundation of China Grant No. 51301091, The Scientific Research Foundation for the Returned Overseas Chinese Scholars, State Education Ministry, and The Natural Science Foundation of Jiangsu Province Grant No. BK20151536.

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